

Access Queries, Forms and Reports — Detailed Lecture Notes

JKSSB Computer Awareness

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30.1 The Access objects and the object sequence

An Access database is built from a small set of **objects**. The **table** stores the actual data. The **query** asks a question and retrieves selected data. The **form** provides a screen for entering and viewing data. The **report** presents data in a formatted, printable layout. These four objects are used together in a natural order: data is stored in a table, selected by a query, entered through a form, and printed by a report. That order — **Table** → **Query** → **Form** → **Report** — is the object sequence this lesson is built around.

Definition (Database object). *A named component of an Access database that performs one job. The main objects are the table, query, form and report.*

Object	Job
Table	Stores the actual data in rows and columns.
Query	Asks a question and retrieves selected data.
Form	Provides an on-screen interface to enter and view data.
Report	Presents data in a formatted, printable layout.

The rest of this lesson uses these terms in one consistent sense. A **table** is the store. A **query** is the question. A **form** is the on-screen entry screen. A **report** is the printed output.

30.2 The table is the source of data

A table stores data in **fields** (columns) and **records** (rows). Every other object draws its data from a table or from a query. A form or report built on a table shows that table's data; built on a query, it shows only the rows the query selected. No data is stored in a query, a form or a report — those objects only *present* what the table holds. This is why the table is called the foundation of the database.

30.3 What a query is

A **query** is a question asked of the database. It retrieves data from one or more tables, usually by a condition. A query does not normally store new data of its own; it shows a *result set* drawn from the tables each time it is opened. Queries can also be used to *change* data; those are called **action queries**, and they are discussed below.

Definition (Query). *An Access object that selects, filters, or acts on records according to the rules the user sets, drawing its data from one or more tables.*

30.4 The Select query: retrieving on a condition

A **Select query** retrieves the records that satisfy a **condition** (also called a *criterion*). It is the most common query type in Access. The condition decides *which rows appear*, and the chosen fields decide *which columns appear*. For example, a Select query on a Students table could show only those students whose **Marks** are greater than 60.

Example 30.4.1. *An item asks: “What is a query that retrieves data on the basis of a condition called?” The query is only retrieving data — it is not changing it — so it is a **Select query**. This is the concept tested by PYQ-39 (Website Operator 2026, query retrieving data on a condition).*

30.5 Building a query: Design view and criteria

Queries are usually built in **Query Design** view, or typed directly in **SQL** view. The design grid is called **QBE** — **Query By Example**. Each column of the grid holds one **field**, and the **Criteria** row holds the condition for that field. A criterion of >60 on the **Marks** field, for instance, keeps only the rows in which marks are above 60.

Definition (QBE (Query By Example)). *The grid-based design view in Access in which a query is built by choosing fields and typing criteria, without writing the SQL by hand.*

Example 30.5.1. *For a Students table, place **Marks** in the field row and >60 in the Criteria row. When the query runs, Access returns only the students who scored more than 60.*

30.6 Action queries: Update, Append and Delete

An **action query** changes the stored data. The three common types are:

- An **Update query** changes the existing values in records.
- An **Append query** adds records from one table to another.
- A **Delete query** removes the records that match a condition.

Because these queries alter the table itself, they are treated with care in Access. The distinction between retrieving data and changing data is the key one: a Select query only reads, while Update, Append and Delete queries write.

Query	What it does
Select	Retrieves records that satisfy a condition; does not change data.
Update	Changes existing values in records.
Append	Adds records from one table to another.
Delete	Removes records that match a condition.

30.7 Select versus action queries

The nearest confusion in this topic is between a Select query and the action queries. A **Select query** **retrieves** data on a condition and leaves the tables unchanged. An **action query** (Update, Append, Delete) **modifies** the stored data. If a question asks which query *changes* records, the answer is an action query; if it asks which query *shows* records that meet a condition, the answer is a Select query. This is the distinction recorded as TRAP-31.

Example 30.7.1. *An item lists Select, Update, Append and Delete queries and asks which one does **not** modify the data. Select is the only one that leaves the tables unchanged, so Select is the answer.*

30.8 Lookup fields

A **lookup** field lets the user *pick a value from a list* instead of typing it. The list usually comes from a table or a query — for example, a list of department names from a Departments table. A lookup is often used for a **foreign key**, so that the value entered matches an existing value in the related table. The benefit is fewer typing errors and more consistent data. A lookup field only *chooses* a value from a list; it does not create a new table or a new query.

Definition (Lookup field). *A field that displays a list of values, drawn from a table or query, so the user can select one instead of typing it.*

30.9 Basic SQL awareness

SQL stands for **Structured Query Language**. It is the standard language for working with relational databases, and Access runs SQL behind the scenes whenever a query is run. A retrieval statement has the general shape `SELECT fields FROM table WHERE condition;`. The **SELECT** keyword names the fields to show, **FROM** names the table, and **WHERE** gives the condition. To retrieve only the students with marks above 60, one would write `SELECT Name FROM Students WHERE Marks > 60;`.

Definition (SQL). *Structured Query Language — the standard language used to define and retrieve data in a relational database.*

30.10 Forms: entering and viewing data

A **form** is an object that provides an on-screen interface for entering, viewing and editing data. A form is usually based on a table or a query. It shows one record at a time (or a few), with clearly labelled fields, and makes data entry easier and safer than typing directly into a table grid. Where the table is the raw store, the form is the friendly screen the user actually works with.

30.11 Reports: printable output

A **report** presents data in a formatted, printable layout. It can group records and show totals and summaries. A report is **read-only**: it is meant for printing and sharing, not for editing data. Like a form, a report can be based on a table or a query. The report is the output object of the database.

Example 30.11.1. *A school can keep all student marks in a table, use a Select query to pick the students who passed, enter new students through a form, and print the pass list as a report. The same data moves, in order, through the Table → Query → Form → Report sequence.*

30.12 Forms versus reports

The nearest confusion between the two presentation objects is worth stating plainly. A **form** is used *on screen* to enter and view data, and it allows editing. A **report** is *read-only* and is used to *print* data and present summaries. A form answers the question “how do I put data in and look at it?”; a report answers “how do I print it?”

Point	Form	Report
Purpose	Enter and view data on screen	Print and present data
Data change	Allows editing	Read-only
Typical use	Data-entry screen	Printed summary

30.13 Exam retrieval and common confusions

Six distinctions clear up most errors on this topic.

1. A **Select** query **retrieves** data on a condition; **Update**, **Append** and **Delete** queries **modify** data (TRAP-31).
2. Data is stored in a **table**, **not** in a query, form or report.
3. A **form** is for on-screen entry and viewing; a **report** is a printable, read-only output.
4. The object sequence is **Table** → **Query** → **Form** → **Report**.
5. **SQL** stands for **Structured Query Language**; `SELECT ... FROM ... WHERE` retrieves data on a condition.
6. A **lookup** field picks a value from a list supplied by a table or query.

Example 30.13.1. *An item states, “A report is used to enter new records into the database.” This is wrong. Entering records is the job of a **form**; a report is **read-only** and is used for printing. The wrong option has swapped the form and the report.*

Exercises

1. Name the four main objects of an Access database and state the job of each.
2. Give the correct object sequence for storing, selecting, entering and printing data.
3. What is a Select query, and how does it differ from an Update query?
4. Name the three common action queries and say what each one does.
5. What is a lookup field, and why is it useful?
6. Expand SQL and write the general form of a retrieval statement.
7. Distinguish a form from a report.

Solutions

1. **Table** stores the data; **Query** asks a question and retrieves selected data; **Form** provides an on-screen interface to enter and view data; **Report** presents data in a formatted, printable layout.
2. **Table** → **Query** → **Form** → **Report**. The table stores the data, the query selects it, the form enters it, and the report prints it.
3. A Select query **retrieves** records that satisfy a condition and leaves the tables unchanged. An Update query **changes** the existing values in records, so it modifies the stored data.
4. **Update** changes existing values in records; **Append** adds records from one table to another; **Delete** removes records that match a condition.
5. A lookup field displays a list of values, drawn from a table or query, so the user selects one instead of typing it. It reduces typing errors and keeps values consistent, and it is often used for a foreign key.
6. SQL is **Structured Query Language**. A retrieval statement has the form **SELECT fields FROM table WHERE condition;** where **SELECT** names the fields, **FROM** names the table, and **WHERE** gives the condition.
7. A **form** is an on-screen interface used to enter and view data, and it allows editing. A **report** is a read-only, formatted output used for printing and summarising data.